

Task 2: Query Understanding (Decomposition Task)

This document contains the structural breakdown (Intent, Tables, Columns, Filters, Joins) for every natural language question in the benchmark dataset, as required for Task 2.

1. List all products

- **Intent:** List all products
- **Tables:** products
- **Columns:** *
- **Filters:** None
- **Joins:** None

2. Get all customers

- **Intent:** Retrieve all customers
- **Tables:** customers
- **Columns:** *
- **Filters:** None
- **Joins:** None

3. Show all orders

- **Intent:** Retrieve all orders
- **Tables:** orders
- **Columns:** *
- **Filters:** None
- **Joins:** None

4. List all employees

- **Intent:** List all employees
- **Tables:** employees
- **Columns:** *
- **Filters:** None
- **Joins:** None

5. Get all offices

- **Intent:** Retrieve all records
- **Tables:** offices
- **Columns:** *
- **Filters:** None
- **Joins:** None

6. Show all product lines

- **Intent:** Retrieve all product lines
- **Tables:** productlines
- **Columns:** *
- **Filters:** None
- **Joins:** None

7. List all payments

- **Intent:** List all records
- **Tables:** payments
- **Columns:** *
- **Filters:** None
- **Joins:** None

8. Get product names and prices

- **Intent:** Retrieve product names and prices
- **Tables:** products
- **Columns:** productName, price
- **Filters:** None
- **Joins:** None

9. Get customer names and cities

- **Intent:** Retrieve customer names and cities
- **Tables:** customers
- **Columns:** customerName, city
- **Filters:** None
- **Joins:** None

10. List employee first and last names

- **Intent:** List employee names
- **Tables:** employees
- **Columns:** firstName, lastName
- **Filters:** None
- **Joins:** None

11. Get all order dates

- **Intent:** Retrieve all order dates
- **Tables:** orders
- **Columns:** orderDate
- **Filters:** None
- **Joins:** None

12. Show product vendor list

- **Intent:** Retrieve list of product vendors

- **Tables:** products
- **Columns:** productVendor
- **Filters:** None
- **Joins:** None

13. Get all product codes

- **Intent:** Retrieve all product codes
- **Tables:** products
- **Columns:** productCode
- **Filters:** None
- **Joins:** None

14. List all countries from offices

- **Intent:** List all countries
- **Tables:** offices
- **Columns:** country
- **Filters:** None
- **Joins:** None

15. Show all order statuses

- **Intent:** Retrieve all order statuses
- **Tables:** orders
- **Columns:** status
- **Filters:** None
- **Joins:** None

16. Get all payment amounts

- **Intent:** Retrieve all payment amounts
- **Tables:** payments
- **Columns:** amount
- **Filters:** None
- **Joins:** None

17. List all job titles

- **Intent:** List all job titles
- **Tables:** employees
- **Columns:** jobTitle
- **Filters:** None
- **Joins:** None

18. Get customer phone numbers

- **Intent:** Retrieve customer phone numbers
- **Tables:** customers

- **Columns:** phone
- **Filters:** None
- **Joins:** None

19. Show product MSRP values

- **Intent:** Retrieve product MSRP values
- **Tables:** products
- **Columns:** MSRP
- **Filters:** None
- **Joins:** None

20. List order numbers

- **Intent:** List order numbers
- **Tables:** orders
- **Columns:** orderNumber
- **Filters:** None
- **Joins:** None

21. Get orders with customer names

- **Intent:** Retrieve orders with customer names
- **Tables:** orders, customers
- **Columns:** orders.orderNumber, customers.customerName
- **Filters:** None
- **Joins:** orders.customerNumber = customers.customerNumber

22. Get employees with office city

- **Intent:** Retrieve employees with their office city
- **Tables:** employees, offices
- **Columns:** employees.*, offices.city
- **Filters:** None
- **Joins:** employees.officeCode = offices.officeCode

23. Get payments with customer names

- **Intent:** Retrieve payments with customer names
- **Tables:** payments, customers
- **Columns:** payments.*, customers.customerName
- **Filters:** None
- **Joins:** payments.customerNumber = customers.customerNumber

24. Get order details with product names

- **Intent:** Retrieve order details with product names
- **Tables:** orderdetails, products
- **Columns:** orderdetails.*, products.productName

- **Filters:** None
- **Joins:** orderdetails.productCode = products.productCode

25. Get products with product line description

- **Intent:** Retrieve products with their product line descriptions
- **Tables:** products, productlines
- **Columns:** products.productName, productlines.textDescription
- **Filters:** None
- **Joins:** products.productLine = productlines.productLine

26. Get customers with sales rep names

- **Intent:** Retrieve customers with their sales representative names
- **Tables:** customers, employees
- **Columns:** customers.customerName, employees.firstName, employees.lastName
- **Filters:** None
- **Joins:** customers.salesRepEmployeeNumber = employees.employeeNumber

27. Get orders with customer city

- **Intent:** Retrieve orders with customer city
- **Tables:** orders, customers
- **Columns:** orders.orderNumber, customers.city
- **Filters:** None
- **Joins:** orders.customerNumber = customers.customerNumber

28. Get employees and their manager

- **Intent:** Retrieve employees and their managers
- **Tables:** employees
- **Columns:** employeeNumber, firstName, lastName, reportsTo
- **Filters:** None
- **Joins:** employees.reportsTo = employees.employeeNumber (self join)

29. Get orderdetails with product vendor

- **Intent:** Retrieve order details with product vendor
- **Tables:** orderdetails, products
- **Columns:** orderdetails.*, products.productVendor
- **Filters:** None
- **Joins:** orderdetails.productCode = products.productCode

30. Get payments with customer country

- **Intent:** Retrieve payments with customer country
- **Tables:** payments, customers
- **Columns:** payments.*, customers.country
- **Filters:** None

- **Joins:** payments.customerNumber = customers.customerNumber

31. Count customers per country

- **Intent:** Count customers per country
- **Tables:** customers
- **Columns:** country, COUNT(customerNumber)
- **Filters:** None
- **Joins:** None

32. Total payments per customer

- **Intent:** Sum total payments per customer
- **Tables:** payments
- **Columns:** customerNumber, amount
- **Filters:** None
- **Joins:** None

33. Number of orders per status

- **Intent:** Count number of orders per status
- **Tables:** orders
- **Columns:** status, COUNT(orderNumber)
- **Filters:** None
- **Joins:** None

34. Products per product line

- **Intent:** Count products per product line
- **Tables:** products
- **Columns:** productLine, COUNT(productCode) AS productCount
- **Filters:** None
- **Joins:** None

35. Employees per office

- **Intent:** Count employees per office
- **Tables:** employees, offices
- **Columns:** officeCode, employeeNumber
- **Filters:** None
- **Joins:** employees.officeCode = offices.officeCode

36. Total stock per product vendor

- **Intent:** Sum total stock quantity
- **Tables:** products
- **Columns:** productVendor, quantityInStock
- **Filters:** None
- **Joins:** None

37. Average buy price per product line

- **Intent:** Calculate average buy price
- **Tables:** products
- **Columns:** productLine, buyPrice
- **Filters:** None
- **Joins:** None

38. Orders per customer

- **Intent:** Count orders per customer
- **Tables:** orders
- **Columns:** customerNumber, COUNT(orderNumber) AS orderCount
- **Filters:** None
- **Joins:** None

39. Max MSRP per product line

- **Intent:** Find maximum MSRP
- **Tables:** products
- **Columns:** productLine, MSRP
- **Filters:** None
- **Joins:** None

40. Min buy price per vendor

- **Intent:** Find minimum buy price per vendor
- **Tables:** vendors, products
- **Columns:** vendorName, buyPrice
- **Filters:** None
- **Joins:** vendors.vendorId = products.vendorId

41. Total number of customers

- **Intent:** Count total customers
- **Tables:** customers
- **Columns:** customerNumber
- **Filters:** None
- **Joins:** None

42. Total number of products

- **Intent:** Count total products
- **Tables:** products
- **Columns:** product_id
- **Filters:** None
- **Joins:** None

43. Total revenue from payments

- **Intent:** Calculate total revenue
- **Tables:** payments
- **Columns:** amount
- **Filters:** None
- **Joins:** None

44. Average product price

- **Intent:** Calculate average
- **Tables:** products
- **Columns:** buyPrice
- **Filters:** None
- **Joins:** None

45. Max payment amount

- **Intent:** Find maximum payment amount
- **Tables:** payments
- **Columns:** amount
- **Filters:** None
- **Joins:** None

46. Min payment amount

- **Intent:** Find minimum payment amount
- **Tables:** payments
- **Columns:** amount
- **Filters:** None
- **Joins:** None

47. Count total orders

- **Intent:** Count total orders
- **Tables:** orders
- **Columns:** orderNumber
- **Filters:** None
- **Joins:** None

48. Total quantity in stock

- **Intent:** Sum total quantity in stock
- **Tables:** products
- **Columns:** quantityInStock
- **Filters:** None
- **Joins:** None

49. Average MSRP

- **Intent:** Calculate average

- **Tables:** products
- **Columns:** MSRP
- **Filters:** None
- **Joins:** None

50. Number of employees

- **Intent:** Count total employees
- **Tables:** employees
- **Columns:** employeeNumber
- **Filters:** None
- **Joins:** None